# Automated Seat-Level Examination Seating Through Paper Aware Constraint Optimisation : A Pratical Framework For Multi - Department Universities

## PHASE 1 REPORT

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*in partial fulfillment for the award of the degree*

*of*

**BACHELOR OF ENGINEERING**
**IN**
**COMPUTER SCIENCE AND ENGINEERING**








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**October 2026**

# BONAFIDE CERTIFICATE

Certified that this Report titled "**AUTOMATED SEAT-LEVEL EXAMINATION SEATING THROUGH PAPER AWARE CONSTRAINT OPTIMISATION : A PRATICAL FRAMEWORK FOR MULTI - DEPARTMENT UNIVERSITIES** " is the bonafide work of **DARSHAN M (230701062), DINISHA R (230701080), JANANI T (230701123), JANANIE S (230701124)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.


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# ABSTRACT

Examination seating arrangement is a complex administrative task when students from multiple departments must be assigned to limited examination halls while satisfying capacity and separation requirements. This paper presents an **Automated Examination Seating Arrangement System** that formulates seating allocation as a constraint optimization problem using the **Google OR-Tools CP-SAT solver**. The system processes Controller of Examinations (COE) data, extracts students from register-number ranges, and maps department-specific course codes to a common **paper_id** to identify students writing the same examination. Halls are modeled using benches and individual seats to enable precise seat-level allocation. Constraints including unique seat assignment, hall capacity, same-paper bench separation, and adjacent-bench restrictions are incorporated to generate valid seating arrangements. The system provides seat-map visualization, student-level seat lookup, and automated arrangement generation.

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# ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.,** for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. Manoranjini,, Ph.D.,** Professor and Head of the Department of Computer Science and Engineering for her guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide**, Mr.Karthikeyan R,** POP, Department of Computer Science and Engineering for his valuable guidance throughout the course of the project. We thank our Project Coordinator, **Dr. S. Madhusudhanan,** Associate Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

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# LIST OF ABBREVATIONS

| | |
|---|---|
| **AI** | Artificial Intelligence |
| **COE** | Controller of Examinations |
| **CP-SAT** | Constraint Programming – Satisfiability |
| **CSV** | Comma-Separated Values |
| **DFD** | Data Flow Diagram |
| **GA** | Genetic Algorithm |
| **OR-TOOLS** | Operations Research Tools |
| **VDM** | Violated Directed Mutation |

# CHAPTER 1
# INTRODUCTION

## 1.1 GENERAL

University examinations involve a large number of students, multiple examination papers, and a limited number of examination halls. Before every examination session, students must be assigned to appropriate halls and individual seats. In many institutions, this process is still carried out using spreadsheets and manual verification. Manual preparation becomes increasingly difficult as the number of students and examination halls increases. It requires repeated checking to ensure that students are assigned correctly, seats are not duplicated, hall capacities are not exceeded, and students writing the same examination paper are appropriately separated. The problem becomes more complicated when students from different departments write the same examination paper using different course codes.

Traditional examination scheduling mainly concentrates on assigning examinations to appropriate time slots and rooms. However, assigning an examination to a room does not determine the exact seat occupied by every student. Seat-level allocation requires consideration of the physical arrangement of benches and the relationship between neighboring seats and benches.

The proposed system therefore considers examination seating as a constraint-based seat allocation problem. The system processes examination data, identifies individual students, maps course codes to a common examination-paper identity, and uses an optimization solver to generate a valid seating arrangement.

The system also provides a configurable benchmark mechanism to test the seating algorithm under different student populations, paper distributions, hall capacities, and cross-department scenarios.

## 1.2 OBJECTIVE

The main objective of the proposed system is to develop an automated examination seating arrangement system capable of assigning students to individual seats while satisfying the required examination constraints.

The specific objectives are:

1. To automate the generation of examination seating arrangements.
2. To reduce the manual effort involved in preparing and verifying seating plans.
3. To process examination data provided in the COE Excel format.
4. To convert register-number ranges into individual student records.

5. To map department-specific course codes to a common paper_id.
6. To ensure that students writing the same examination paper are appropriately separated.
7. To represent examination halls at the hall, bench, and seat levels.
8. To prevent multiple students from being assigned to the same physical seat.
9. To ensure that the number of students assigned to a hall does not exceed its available capacity.
10. To consider same-bench and adjacent-bench restrictions.
11. To use the CP-SAT optimization technique for constraint-aware seat allocation.
12. To provide a seat map and student-level seat lookup.
13. To evaluate the system based on constraint satisfaction, seat utilization, execution time, and scalability.

## 1.3 EXISTING SYSTEM

In the existing system, examination seating arrangements are commonly prepared using spreadsheets or manual allocation methods. The examination staff generally collect student information, examination details, hall capacities, and register-number ranges and then manually assign students to available seats.

The existing examination seating process generally involves collecting examination data, identifying students and their respective courses, assigning students to available examination halls, and manually allocating seats. The allocated seating arrangement must then be checked to ensure that hall capacity is not exceeded and that students writing the same examination paper are not seated together. Finally, the entire seating arrangement is rechecked for errors before preparing the final seating list.

When the number of students and halls increases, the manual process becomes time-consuming and difficult to maintain. It may also result in incorrect assignments, unused seats, duplicate assignments, or repeated manual checking.

Another limitation is that conventional examination timetabling systems generally focus on examination periods and rooms rather than individual seats. Therefore, they do not necessarily model the physical relationships between benches and seats.

### 1.3.1 Limitations of Existing System

The existing examination seating system has several limitations, including a high dependency on manual work and significant time consumption, especially when handling a large number of students. Managing multiple examination halls can also become difficult, and

repeated verification is required to ensure the correctness of the seating arrangement. There is also a possibility of duplicate or incorrect seat assignments.

Verifying whether students writing the same examination paper are properly separated can be challenging when done manually. In addition, relying only on course codes for separation may lead to errors when students from different departments use different course codes for the same examination paper. The existing process may also fail to consider the physical relationships between neighboring benches, making proper seat allocation more difficult. Furthermore, locating the exact seat assigned to an individual student can be inconvenient and time-consuming.

## 1.4 PROPOSED SYSTEM

The proposed system automates examination seating by modelling it as a seat-level constraint optimization problem.

The system accepts examination information from the COE Excel format. Register-number ranges are processed to generate individual student records. Course codes are then mapped to a common paper_id, allowing students from different departments who write the same examination paper to be treated as a single examination group.

The examination hall is represented using the structure:

Hall → Bench → Seat

In the current model, each bench contains two seats, identified as Seat A and Seat B. This representation enables the system to apply both seat-level and bench-level constraints.

The CP-SAT solver from Google OR-Tools is used as the core allocation engine. Instead of first generating an arrangement and then checking whether it violates constraints, the required constraints are directly included in the optimization model.

The proposed system applies several constraints to ensure that the generated examination seating arrangement is valid and conflict-free. Each student is assigned exactly one seat, while each physical seat can be occupied by at most one student.

The number of students assigned to a hall must not exceed its available seating capacity. Students writing the same examination paper are not allowed to occupy the same bench, and they are also prevented from being assigned to prohibited adjacent benches. These constraints help maintain proper student separation and ensure an organized and reliable examination seating arrangement.

After a feasible solution is generated, the allocation is converted into a seating plan containing the student's hall, bench number, and seat

position. The arrangement can then be displayed using a seat map and used for student seat lookup.

The proposed system therefore provides an end-to-end process from examination data upload → data processing → paper mapping → seat allocation → seat map → student lookup.

# 2. LITERATURE SURVEY

Various researchers have proposed mathematical programming, metaheuristic, hybrid, graph-based, and AI-based approaches for examination timetabling and seating problems.

Several studies focus on examination timetabling where examinations are assigned to appropriate periods and rooms. However, their decision variables generally operate at the timetable or room level rather than assigning individual students to physical seats.

Kolawole et al. proposed the use of Simulated Annealing for examination seating arrangements. Their work focused on preventing undesirable adjacency between students taking the same course. The study demonstrated that examination seating can be treated as a combinatorial optimization problem. However, it used a relatively small-scale setting compared with the configurable and multi-department approach of the proposed system.

Reshamwala, Chougule, and Pathan proposed SeatMatrix, a cloud-based examination management system that provides real-time seat mapping. The study demonstrates the usefulness of cloud technologies for examination information management and seating

visualization. However, it mainly focuses on management and visualization rather than constraint-based optimization for generating valid seating arrangements.

Jia et al. proposed a multi-strategy examination room seating approach that combines evolutionary algorithms with a shuffled divide-and-conquer strategy. Their approach demonstrates the application of evolutionary optimization to improve the seating search process. The proposed system differs by explicitly modelling hard constraints involving student and bench separation.

Mankar et al. developed an automated examination seating system using web technologies. Their work demonstrates that automation can reduce administrative effort. However, web-based implementation alone does not solve the underlying optimization problem. The proposed system combines an automated interface with a dedicated constraint optimization engine.

Kibona et al. proposed an examination seating optimization method using graph coloring and combinatorial design. Their study demonstrated the potential of graph-based optimization for large-scale examination seating. However, the proposed system specifically models individual seats, benches, and hard separation constraints.

Other recent studies have applied Genetic Algorithms and data-driven optimization to examination timetabling. These studies reinforce the importance of optimization techniques for examination scheduling but do not explicitly combine individual seat allocation with physical bench and seat relationships.

| REFERENCE | METHOD | SEAT-LEVEL ALLOCATION | PAPER MAPPING | CROSS DEPARTMENT |
|---|---|---|---|---|
| [2]–[14], [20]–[22] | Optimization / Metaheuristic | No | No | No |
| [15] | Simulated annealing | Yes | No | No |
| [16] | Cloud-Based Seat Mapping | Yes | No | No |
| [17] | Evolutionary Approach | Yes | No | No |
| [18] | Web-Based System | Yes | No | No |
| [19] | Graph Coloring | Yes | No | No |
| Proposed System | CP-SAT | Yes | Yes | Yes |

**Table 2.1 Comparison of Recent Examination Seating and Timetabling Studies**

## 2.1 COMPARISON AND DISCUSSION

The literature indicates that examination scheduling has been extensively studied using mathematical programming, metaheuristic algorithms, graph-based approaches, and web-based systems. However, there remains a need for a system that combines seat-level allocation, physical seating constraints, and examination-paper

identity across departments.

A major issue in multi-department examinations is that students may use different course codes even though they are writing the same examination paper. Therefore, simply comparing course codes is insufficient for enforcing separation constraints.

The proposed system addresses this issue by converting different department-specific course codes into a common paper_id. Students associated with the same paper are consequently treated as members of the same conflict group.

The proposed approach also differs from simpler allocation techniques because the constraints are considered collectively during the optimization process. This allows the system to determine whether a complete valid seating arrangement exists rather than simply assigning seats sequentially.

The literature comparison highlights the importance of integrating seat-level allocation, paper-level identification, cross-department examination handling, bench-level restrictions, adjacent-bench restrictions, and constraint-based optimization into a single examination seating system. Combining these features enables the system to accurately identify students writing the same examination

paper, consider the physical arrangement of seats and benches, and automatically generate a valid seating arrangement while satisfying the required constraints.

## 2.1 CONCLUSION

The literature survey shows that several optimization techniques have been successfully applied to examination scheduling and seating. Simulated Annealing, evolutionary algorithms, graph coloring, Genetic Algorithms, cloud-based systems, and web-based systems have each addressed different aspects of the problem.

However, most existing approaches do not combine physical seat and bench relationships with a common examination-paper identity across departments. The proposed system addresses this research gap using CP-SAT-based seat-level allocation, course-code-to-paper_id mapping, configurable adjacency constraints, and reproducible benchmark instances.

# 3. SYSTEM DESIGN

## 3.1 SYSTEM ARCHITECTURE

The proposed system follows a modular architecture in which examination data passes through several processing and allocation stages. The major components of the architecture are: COE Data Upload, Data Processing, Student Record Generation, Course Code to Paper ID Mapping, Hall and Seat Configuration, Constraint Model Construction, CP-SAT Optimization Engine, Seat Allocation, Seat Map Generation, Student Seat Lookup.

The input to the system is examination data provided in the COE Excel format. Register-number ranges are processed to obtain individual student records. The course codes are then mapped to their corresponding paper_id.

The student information, paper information, hall configuration, and seat structure are combined to construct the constraint model. The CP-SAT solver searches for a seating arrangement satisfying the defined hard constraints.

Once a feasible arrangement is generated, the assigned hall, bench,

and seat information is presented through the seat-map interface. The generated information can also be used for student-level seat lookup and examination-office reporting.

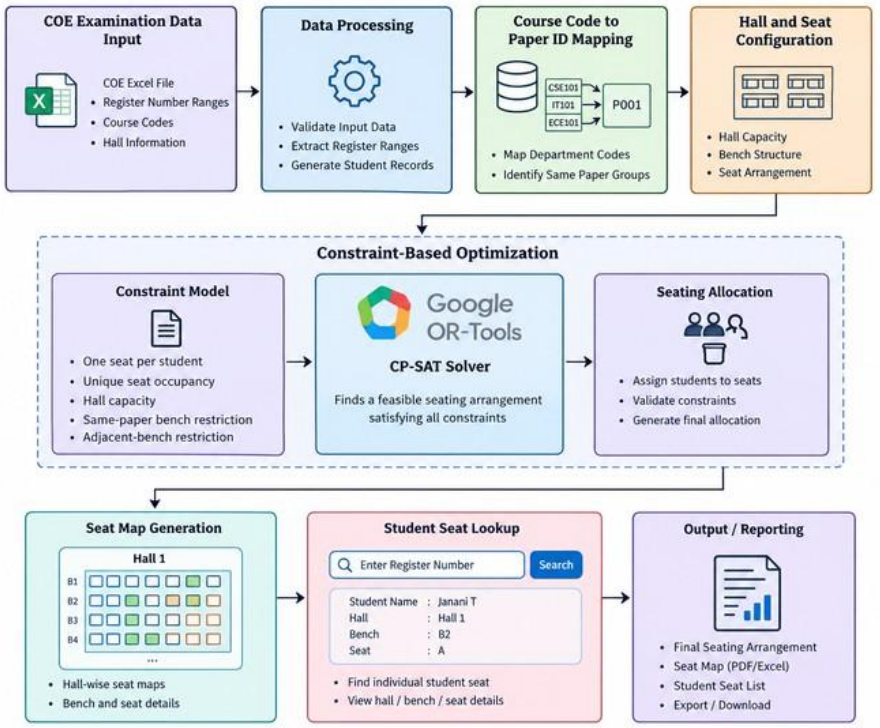



**Figure 3.1 – System Architecture**

The system architecture represents the overall workflow of the Automated Examination Seating Arrangement system. It starts with COE examination data, followed by data processing, Paper ID mapping, and hall and seat configuration. The processed data is then optimized using CP-SAT constraints to generate the final seating

arrangement, seat maps, and student seat details.

## 3.2 SYSTEM REQUIREMENTS

The software requirements for implementing the proposed system include:

| SOFTWARE | PURPOSE |
|---|---|
| Operating System | Provides the execution environment |
| Python | Implementation of seating allocation and optimization logic |
| Google OR-Tools | Provides the CP-SAT constraint solver |
| Excel / Spreadsheet Support | Processing COE examination data |
| JSON | Storage of benchmark specifications |
| CSV | Storage of course and paper mapping information |
| Web Browser | Accessing the system interface |
| Code Editor / IDE | Development and maintenance |

**Table 3.1 Software Requirements**

The core optimization engine uses Google OR-Tools CP-SAT to solve the seat allocation problem while simultaneously considering the defined constraints.

### 3.2.1 HARDWARE REQUIREMENTS

The system does not require specialized hardware because the seating

allocation is performed computationally.

| COMPONENT | REQUIREMENT |
|---|---|
| Processor | Dual-core processor or higher |
| RAM | Minimum 4 GB |
| Storage | Minimum 10 GB free space |
| Display | Standard monitor |
| Input | Keyboard and mouse |
| Network | Required if the system is deployed as a web-based application |

**Table 3.2 Hardware Requirements**

For larger examination instances, higher processing capability and memory can improve execution performance.

### 3.2.2 DATASET REQUIREMENTS

The system requires examination information containing student and examination details.

The primary input is the COE Excel hall-plan format, which contains information such as:

- Class information.
- Course codes.

- Register-number ranges.
- Examination-related student information.
- Hall information.
- Seating capacity.

Register-number ranges are expanded into individual student records before the allocation process begins. For example, a range such as 22CS001–22CS030 is converted into the corresponding individual student entries.

The system also requires course-code-to-paper mapping information. This mapping is necessary because students from different departments may use different course codes while writing the same examination paper.

For algorithm testing, benchmark datasets can contain different:

- Student populations.
- Number of examination papers.
- Number of halls.
- Hall capacities.
- Paper distributions.
- Cross-department/shared-paper scenarios.

The benchmark generator supports balanced, Zipfian, stress-dominant, and cross-department distributions and uses fixed random seeds to make experiments reproducible.

### 3.2.3 DEPLOYMENT AND SCALING

The proposed system can be deployed in an examination administration environment where examination data is uploaded and processed to generate seating arrangements. The system is designed to support different examination sizes. Benchmark experiments range from small sessions involving 60 students to a campus-wide scenario involving 1,000 students.

The scalability of the system depends on the number of students, seats, halls, and constraints. In the experiments, solver runtime increased from 83.26 ms for a 60-student instance to 15.93 seconds for a 1,000-student instance.

The system can therefore be extended to larger environments by improving the optimization configuration, supporting additional hall geometries, and evaluating further algorithms as proposed in the future work.

## 3.3 SYSTEM DESIGN

The system design describes the internal flow of information from examination data input to the final seating arrangement.

The overall process consists of the following stages:

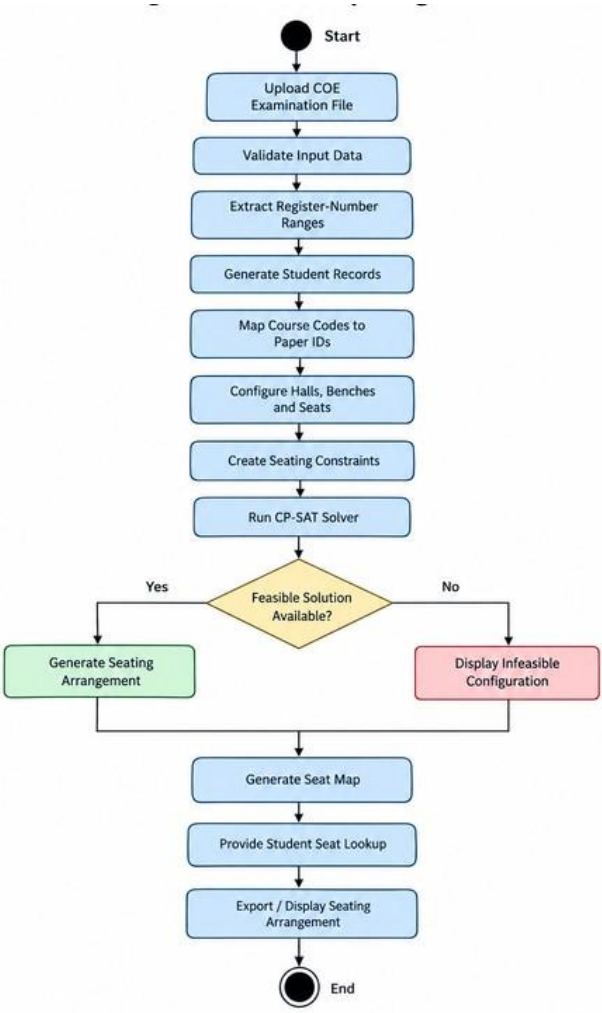
1. Examination data upload.
2. Register-number range processing.
3. Student record generation.
4. Course-code-to-paper-ID mapping.
5. Hall and seat representation.
6. Constraint formulation.
7. CP-SAT-based allocation.
8. Validation of the generated solution.
9. Seat-map generation.
10. Student seat lookup and export.

The system ensures that each student receives exactly one seat and that no physical seat is assigned to more than one student. It also considers hall capacity, same-paper restrictions, and adjacent-bench constraints.

### 3.3.1 ACTIVITY DIAGRAM

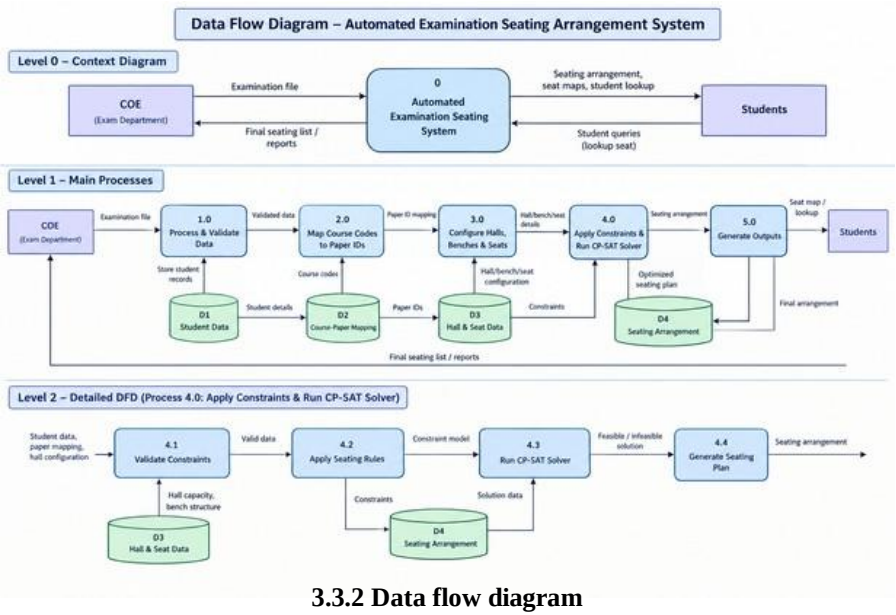
The activity diagram shows the sequence of the system from COE file upload to final seating allocation. It includes data validation, student record generation, Paper ID mapping, hall configuration, constraint creation, and CP-SAT optimization. Based on the solution,

the system generates the seat map and student seat details or displays an infeasible configuration.




**3.3.1 Activity diagram**

## 3.3.2 DATA FLOW DIAGRAM




**3.3.2 Data flow diagram**

The Data Flow Diagram shows how examination data moves through the system, from input and processing to constraint-based seating allocation and the generation of seat maps and final seating reports.

# 4. PROJECT DESCRIPTION

The proposed Automated Examination Seating Arrangement System is designed to automatically generate an organized and constraint-satisfying seating arrangement for university examinations. The system processes examination data, identifies students and their respective examination papers, maps different course codes representing the same paper to a common paper ID, and represents the physical arrangement of halls, benches, and seats. It then uses the Google OR-Tools CP-SAT solver to allocate each student to a suitable seat while satisfying examination seating constraints such as hall capacity, unique seat allocation, same-paper separation, and adjacent-bench restrictions. The generated seating arrangement can then be viewed through seat maps and individual student seat lookup.

## 4.1 METHODOLOGIES

### 4.1.1 Problem Formulation

The examination seating problem is formulated as a seat-level constraint optimization problem. The system considers students, examination papers, halls, benches, and seats and determines a suitable seat for every student without violating the defined examination rules. Each student must receive exactly one seat, while each physical seat can be assigned to at most one student. Hall

capacity, paper identity, and physical bench relationships are also considered during allocation.

### 4.1.2 Data Preprocessing

The examination data provided by the Controller of Examinations is processed before generating the seating arrangement. The system validates the input information and extracts student details from the given register-number ranges. These processed records are then used as input for the seating allocation process.

### 4.1.3 Course Code to Paper ID Mapping

Students from different departments may have different course codes while writing the same examination paper. Therefore, the system maps such course codes to a common paper_id. This paper ID is used for applying separation constraints, ensuring that students writing the same actual paper are treated as the same group even when their department-specific course codes differ.

### 4.1.4 Hall and Seat Representation

Each examination hall is represented according to its physical seating structure. The system considers the available benches and seats within each hall. In the proposed arrangement, each bench contains

two seating positions, identified as A and B. The seating structure is represented as:

Hall → Bench → Seat

This representation allows the system to perform allocation at the individual seat level while applying restrictions at the bench and adjacent-bench levels.

### 4.1.5 Constraint Formulation

The seating arrangement is generated by applying a set of constraints to ensure that every student receives a valid seat and that examination rules are satisfied.

**1. Student Assignment Constraint**

Each student must be assigned exactly one seat.

$$\sum_{h,b,k} x_{s,h,b,k} = 1 \qquad \forall s$$

where h,b,k∑xs,h,b,k=1 ,if student is assigned to seat of bench in hall , and 0 otherwise.

**2. Seat Occupancy Constraint**

Each physical seat can be assigned to at most one student.

$$\sum_{s} x_{s,h,b,k} \leq 1 \qquad \forall h, b, k$$

This prevents two students from being assigned to the same physical seat.

**3. Hall Capacity Constraint**

The number of students assigned to a hall must not exceed the available seating capacity. This ensures that the seating arrangement remains within the capacity of each examination hall.

**4. Same-Paper Bench Constraint**

Students writing the same examination paper must not occupy the two seats of the same bench. The system uses paper_id rather than only the department-specific course code, so this restriction also works for students from different departments writing the same paper.

**5. Adjacent-Bench Constraint**

Students writing the same paper must also be prevented from occupying benches that are configured as prohibited neighboring

benches. The physical seating arrangement is represented as a graph:

$$G=(V, E)$$

where $V$ represents the benches and $E$ represents the adjacency relationships between benches.

## 4.2 MODULES

The system is divided into multiple modules, with each module performing a specific function in the examination seating process. These modules work together to process the examination data, generate the seating arrangement, validate the constraints, and provide the final seating information to the examination administrator.

### 4.2.1 Input Data Processing Module

This module handles the examination data provided by the Controller of Examinations. It validates the required information and processes register-number ranges to generate individual student records. The processed student information is then passed to the subsequent modules.

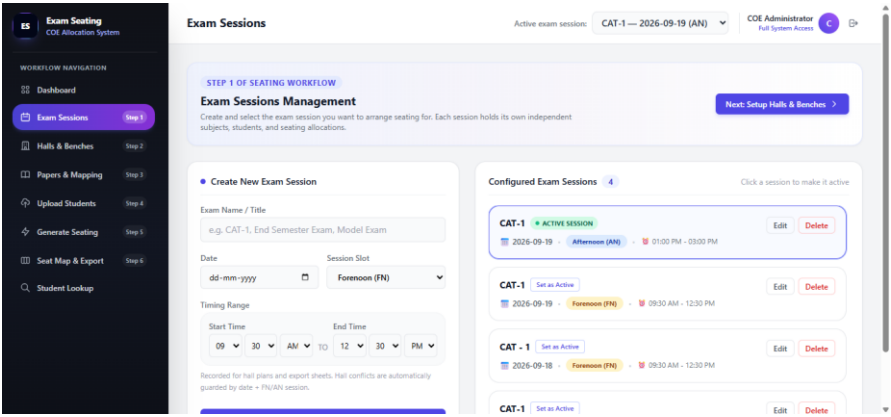



Image 4.2.1 Exam session creation

### 4.2.2 Course Code to Paper ID Mapping Module

This module identifies different course codes that correspond to the same examination paper and maps them to a common paper ID. This ensures that the same-paper separation rules are correctly applied to students from different departments.

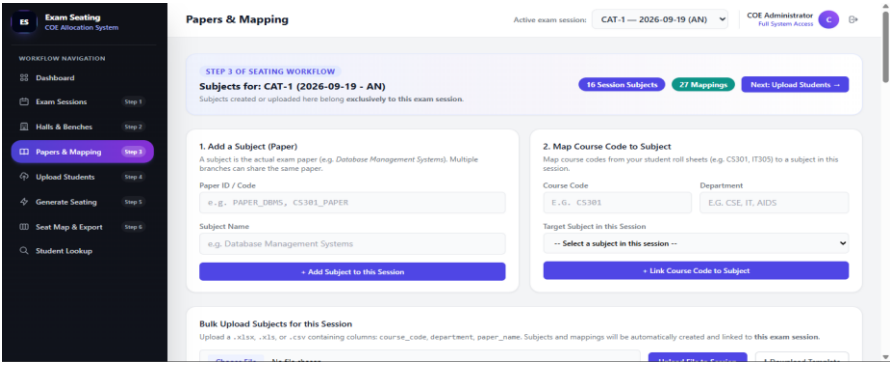



Image 4.2.2. Paper Mapping Module

### 4.2.3 Hall and Seat Management Module

This module manages the examination halls and their seating structures. It stores information about halls, benches, seat positions, hall capacity, and relationships between neighboring benches. The Hall → Bench → Seat structure is used for individual seat allocation.

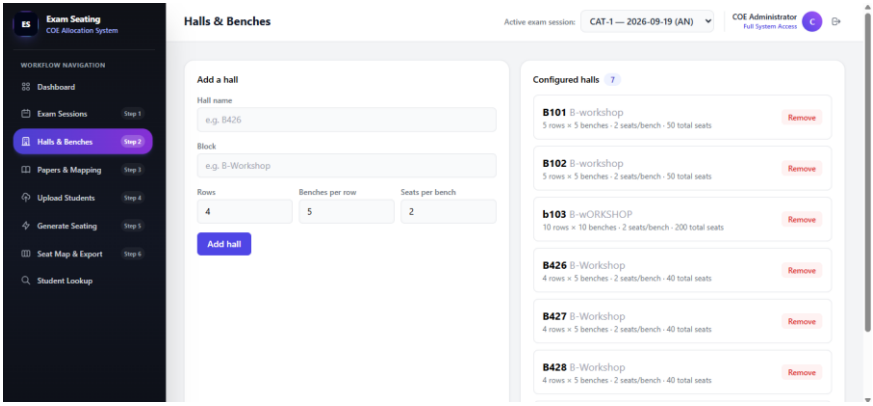



Image 4.2.3. Hall Creation Page

### 4.2.4 Upload Student Module

This is the main module responsible for generating the examination seating arrangement. It receives the processed student, paper, hall, and seat information and uses the CP-SAT solver to assign students to appropriate seats while satisfying the defined constraints.

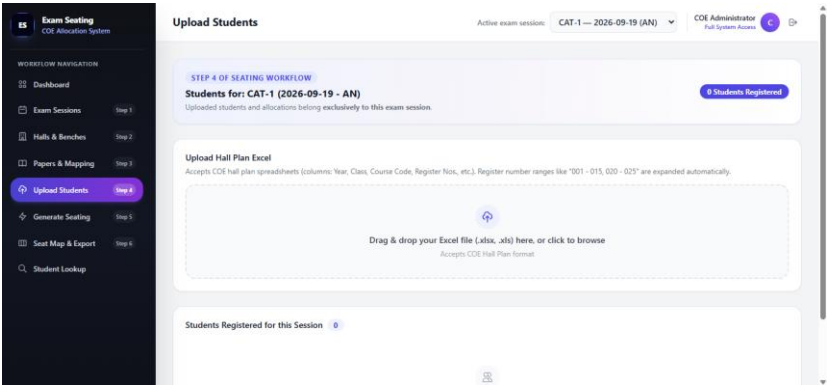



Image 4.2.4.Upload Student

### 4.2.5 Constraint Validation Module

This module verifies whether the generated seating arrangement satisfies all required constraints. It checks for unassigned students, duplicate seat assignments, hall-capacity violations, same-paper students sharing a bench, and prohibited adjacent-bench placements.

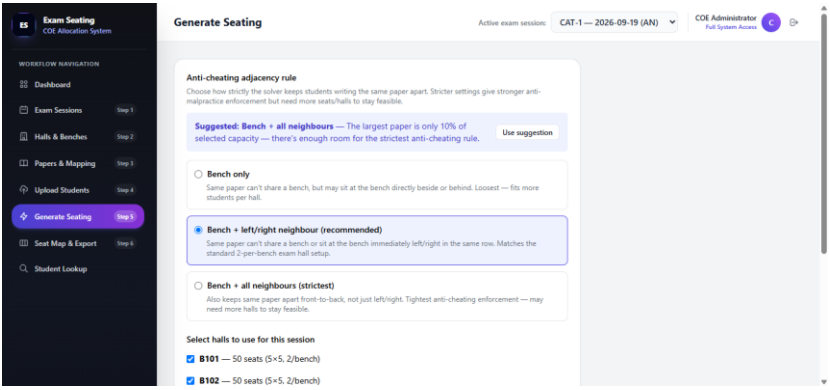



Image 4.2.5. Constraint Validation

### 4.2.6 Seat Map Generation Module

After a valid seating arrangement is generated, this module converts the allocation data into a hall-wise seat map. The seat map allows examination administrators to easily view the arrangement and identify the students assigned to individual seats.

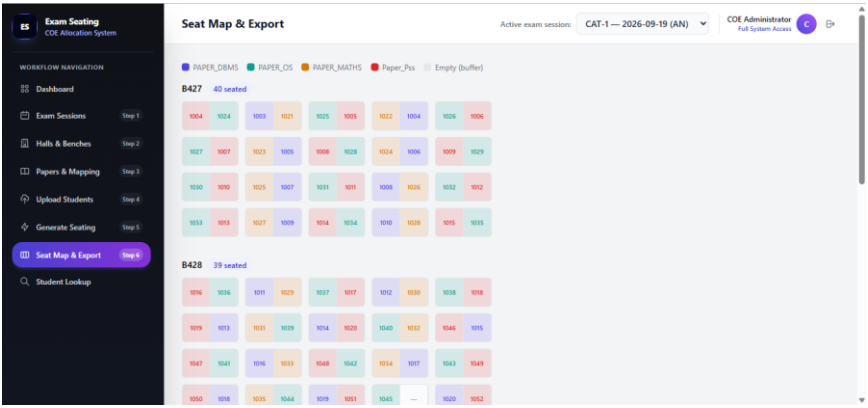



Image 4.2.6. SeatMap Generation

# 5. IMPLEMENTATION AND RESULT DISCUSSION

## 5.1 IMPLEMENTATION RESULTS

The proposed examination seating system was implemented using Google OR-Tools CP-SAT to perform constraint-based seat allocation. The implementation was evaluated using different benchmark instances to study the behaviour of the system under varying numbers of students, examination papers, halls, and seating capacities. The results were analyzed based on constraint satisfaction, feasibility, execution time, seat utilization, and scalability.

### 5.1.1 BENCHMARK IMPLEMENTATION AND TESTING

The system was tested using benchmark datasets designed to represent different university examination scenarios. The datasets varied in terms of the number of students, number of examination papers, available halls, hall capacities, and distribution of students among papers.

The evaluation included balanced, Zipfian, stress-dominant, cross-department, large-scale, campus-wide, and full-grid strict scenarios. The smallest benchmark contained 60 students, while the largest

contained 1,000 students, allowing the system to be evaluated under both small and large examination conditions.

For each benchmark, the student information was processed and different course codes representing the same examination were mapped to a common paper_id. The hall configuration was represented at the bench and seat level before being provided to the CP-SAT solver.

The solver was configured with a maximum execution time of 15 seconds and 8 workers for the experimental evaluation.The implementation was evaluated using important performance measures such as constraint violations, seat utilization, number of unassigned students, number of halls used, execution time, and solution feasibility. These measures help determine whether the generated seating arrangement satisfies the required examination constraints and whether the approach can handle larger datasets efficiently.

### 5.1.2 CP-SAT ALLOCATION RESULTS

The experimental results showed that the CP-SAT solver successfully generated complete seating arrangements for feasible benchmark instances while satisfying the specified seating constraints. For the smaller balanced instance containing 60 students, the solver obtained

an optimal solution in approximately 83.26 milliseconds with 75% seat utilization.

Several benchmark instances were correctly identified as infeasible. This occurred mainly when a large number of students writing the same paper could not be sufficiently separated because of the physical bench and adjacency restrictions. Therefore, even when the total number of available seats was greater than the number of students, a valid seating arrangement was not always possible.

For the largest campus-wide benchmark containing 1,000 students and 15 halls, the CP-SAT solver generated a feasible seating arrangement in approximately 15.93 seconds, achieving 83.33% seat utilization. Although optimality was not established within the specified time limit, the solver was able to produce a valid complete allocation for the large examination scenario.

The scalability results indicate that execution time increases as the number of students, seats, and constraints increases. Nevertheless, the successful allocation of the 1,000-student benchmark demonstrates that the proposed approach can be applied to relatively large university examination scenarios.The results also show that physical seating structure and paper distribution are important factors in determining whether an examination seating problem is feasible.

## 5.2 DISCUSSION

The experimental results demonstrate that the proposed CP-SAT based examination seating system can generate complete seat-level allocations when the available hall configuration satisfies the required constraints. The solver ensures that students are assigned unique seats while maintaining hall capacity, same-paper bench restrictions, and adjacent-bench restrictions.

The results also show that having sufficient total seating capacity alone does not guarantee a feasible arrangement. When a large number of students write the same examination paper, the physical arrangement of benches and adjacency restrictions may make the problem infeasible. This highlights the importance of considering the actual hall and bench structure during examination seating allocation.

When compared with the greedy allocation method, the greedy approach produced results faster but was able to generate complete allocations for only 2 out of 7 benchmark instances, whereas CP-SAT produced complete solutions for 3 out of 7 instances and correctly identified infeasible cases. For the large 1,000-student instance, CP-SAT generated a feasible solution in approximately 15.93 seconds.

Overall, the results indicate that the proposed approach provides a more reliable method for examination seating allocation by combining paper-aware mapping, physical seat-level constraints, and constraint optimization.

# 6. CONCLUSION AND FUTURE WORK

This chapter summarizes the outcomes of the proposed automated examination seating system and discusses possible improvements that can be incorporated in the future. The proposed approach combines paper-aware mapping with seat-level constraint optimization to generate examination seating arrangements. The effectiveness of the system is evaluated using different benchmark scenarios, including large-scale examination instances.

## 6.1 CONCLUSION

The proposed system provides an automated approach for generating seat-level examination seating arrangements using Google OR-Tools CP-SAT. Unlike traditional approaches that mainly focus on examination timetabling or room allocation, the proposed system considers the exact hall, bench, and seat assigned to each student.

A major feature of the system is the use of a common paper_id to identify students writing the same examination paper even when different course codes are used across departments. This allows the system to correctly enforce same-paper bench and adjacent-bench restrictions across departmental boundaries.

The experimental evaluation demonstrates that CP-SAT can generate complete seating arrangements for feasible examination scenarios and can also correctly identify situations where the available physical seating structure makes an allocation infeasible.

The successful generation of a feasible arrangement for a 1,000-student campus-wide instance demonstrates the scalability of the proposed approach. Overall, the system reduces the complexity and manual effort involved in examination seating preparation while providing a structured, constraint-aware, and practical solution for university examination management.

## 6.2 FUTURE WORK

The proposed system provides an automated approach for generating seat-level examination seating arrangements using Google OR-Tools CP-SAT. Unlike traditional approaches that mainly focus on examination timetabling or room allocation, the proposed system considers the exact hall, bench, and seat assigned to each student.

A major feature of the system is the use of a common paper_id to identify students writing the same examination paper even when different course codes are used across departments. This allows the system to correctly enforce same-paper bench and adjacent-bench

restrictions across departmental boundaries.

The experimental evaluation demonstrates that CP-SAT can generate complete seating arrangements for feasible examination scenarios and can also correctly identify situations where the available physical seating structure makes an allocation infeasible. The successful generation of a feasible arrangement for a 1,000-student campus-wide instance demonstrates the scalability of the proposed approach.

Overall, the system reduces the complexity and manual effort involved in examination seating preparation while providing a structured, constraint-aware, and practical solution for university examination management.

# REFERENCES

[1] E. S. K. Siew, S. N. Sze, S. L. Goh, G. Kendall, N. R. Sabar, and S. Abdullah, “A survey of solution methodologies for exam timetabling problems,” IEEE Access, vol. 12, pp. 41479–41498, 2024, doi: 10.1109/ACCESS.2024.3378054.

[2] I. G. A. Premananda, A. Tjahyanto, and A. Muklason, “Timetabling problems and the effort toward generic algorithms: A comprehensive survey,” IEEE Access, vol. 12, pp. 143854–143868, 2024, doi: 10.1109/ACCESS.2024.3463721.

[3] R. Nand, E. Reddy, K. Chaudhary, and B. Sharma, “Preference-based stepping ahead firefly algorithm for solving real-world uncapacitated examination timetabling problem,” IEEE Access, vol. 12, pp. 24685–24699, 2024, doi: 10.1109/ACCESS.2024.3365734.

[4] A. Muklason, E. J. Pratama, and I. G. A. Premananda, “Generic university examination timetabling system with steepest-ascent hill climbing hyper-heuristic algorithm,” Procedia Computer Science, vol. 234, pp. 584–591, 2024.

[5] T. Laisupannawong and S. Sumetthapiwat, “An integer linear programming model for the examination timetabling problem: A case study of a university in Thailand,” Advances in Operations Research, vol. 2024, pp. 1–17, 2024.

[6] Z. Kamisli Ozturk, H. S. Gundogan, E. Mumyakmaz, and T. Kececioglu, “Exam scheduling under pandemic conditions: A mathematical model and decision support system,” Technological Forecasting and Social Change, vol. 208, p. 123687, 2024.

[7] I. Lulu, H. Alowais, A. Turky, S. Harous, and A. Hussain, “Efficient solution for exam timetabling problem: A case study of the University of Sharjah,” in Proc. 17th Int. Conf. Developments in eSystems Engineering (DeSE), 2024, pp. 101–106.

[8] S. A. Wardah, A. Muklason, and I. G. A. Premananda, “Efficient hybrid iterated local search with threshold acceptance algorithm for addressing real world examination timetabling problem,” in Proc. 7th Int. Seminar on Research of Information Technology and Intelligent Systems (ISRITI), 2024, pp. 296–302.

[9] A. Dimitsas, C. Gogos, C. Valouxis, V. Nastos, and P. Alefragis, “A proven optimal result for a benchmark instance of the uncapacitated examination timetabling problem,” Journal of

Scheduling, vol. 28, pp. 183–194, 2025.

[10] B. Bassimir and R. Wanka, “On the computation of robust examination timetables: Methods and experimental results,” Journal of Scheduling, vol. 28, pp. 159–181, 2025.

[11] D. Van Bulck, D. Goossens, and A. Schaerf, “Multi-neighbourhood simulated annealing for the ITC-2007 capacitated examination timetabling problem,” Journal of Scheduling, vol. 28, no. 2, pp. 217–232, 2025.

[12] E. S. K. Siew, S. N. Sze, and S. L. Goh, “Optimizing decentralized exam timetabling with a discrete whale optimization algorithm,” International Journal of Advanced Computer Science and Applications, vol. 16, no. 1, pp. 257–265, 2025.

[13] D. M. K. Dewi and A. Musdholifah, “Final assignment exam scheduling optimization using genetic algorithms with tournament selection techniques and violated directed mutation (VDM),” Science, Technology, and Communication Journal, vol. 5, no. 2, pp. 67–80, 2025.

[14] A. Ghaffar, I. U. Din, A. Tariq, and M. H. Zafar, “Hybridization and artificial intelligence in optimizing university examination timetabling problem: A systematic review,” Review of Education, vol. 13, no. 2, e70071, 2025.

[15] A. S. Kolawole, M. D. Olabisi, and S. O. Femi, “Optimisation of students' examination seating arrangements using simulated annealing,” International Journal of Recent Advances in Multidisciplinary Topics, vol. 6, no. 7, pp. 11–14, Jul. 2025.

[16] K. Reshamwala, S. Chougule, and M. S. Pathan, “SeatMatrix: A cloud-based intelligent examination management system with real-time seat mapping,” Journal of Information and Communications Technology: Algorithms, Systems and Applications, vol. 2, no. 2, p. 26307, 2026.

[17] L. Jia, Q. Li, G. Hou, B. Guo, K. Zhao, C. Yao, and Z. Liu, “A multi-strategy approach for examination room seating arrangement in graduate admissions of petroleum universities: Integrating evolutionary algorithms and shuffled divide-and-conquer strategies,” in Proc. 2025 4th Int. Conf. Artificial Intelligence and Education, 2026, pp. 707–712.

[18] O. Mankar, G. Savji, V. Kolte, and S. Dalave, “Automated MCQ examination seating arrangement system using web technologies,” International Journal of Emerging Technologies and Innovative Research, vol. 13, no. 1, pp. a294–a301, Jan. 2026.

[19] I. E. Kibona, J. January, H. Matimbwa, E. Nchia, W. Matungwa, and S. A. H. Vuai, “Examination seating optimization using graph coloring and combinatorial design,” Array, vol. 30, p. 100897, 2026.

[20] K. Chebil, M. Khemakhem, A. Hammad, and M. Alanazi, “Examination timetabling with independent resource pools: A multi-stage genetic algorithm approach,” Computer Science and Information Systems, vol. 23, no. 3, pp. 1027–1052, 2026.

[21] X. Lai, X. Lu, and B. Jin, “Data-driven university examination timetabling with consideration of student stress,” Computers & Operations Research, vol. 192, p. 107496, 2026.

[22] K. Akgol, M. S. Gor, and Ö. F. Öztürk, “A genetic algorithm application for university exam timetabling: A real-world study,” Journal of Engineering Research, 2026.